Hyperspectral Image Denoising Using a 3-D Attention Denoising Network

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***Abstract*— Hyperspectral image (HSI) denoising plays an important role in image quality improvement and related applications. Convolutional neural network (CNN)-based image denoising methods have been predominant due to advances made in the field of deep learning in recent years. Spatial and spectral information are crucial to HIS denoising, along with their correlations. However, existing methods fail to consider the global dependence and correlation between spatial and spectral information. Accordingly, in this article, we propose a novel dual-attention denoising network to overcome these limitations. We design two parallel branches to process the spatial and spectral information separately. The position attention module is applied to the spatial branch to formulate the interdependencies on the feature map, while the channel attention module is applied to the spectral branch to simulate the spectral correlation before the two branches are combined. A multiscale structure is also employed to extract and fuse the multiscale features following the fusion of spatial and spectral information. Experimental results on simulated and real data substantiate the superiority of our method both visually and quantitatively when compared with state-of-the-art methods.**

***Index Terms*— Atrous convolution, convolutional neural network (CNN), hyperspectral image (HSI) denoising, multiscale structure, self-attention.**

I. Introduction

HYPERSPECTRAL images (HSIs) have been widely applied in the field of remote sensing to tackle various problems, including classification ([1], [2]), spectral unmixing [3], and target detection [4]. Due to the presence of detailed and abundant spectral information contained in the dozens to hundreds of bands of HSIs, the spectral features of different objects obtained from HSIs are better able to be distinguished compared with those derived from multispectral images. However, the multidetector imaging system of HSIs introduces complicated noise, including random noise and structural stripe noise; this mixed-noise situation significantly affects subsequent information extraction and comprehension. Accordingly, it is crucial to implement a pre-processing denoising step prior to HSI data interpretation.

Numerous noise removal methods for HSIs have been proposed over the past decades. Conventional image denoising methods are primarily based on band-wise modeling; these include the block-matching and 3-D filtering (BM3-D) algorithm, based on enhanced sparse representation [5], and the weighted nuclear norm minimization (WNNM) algorithm, which exploits the image's nonlocal self-similarity [6]. However, such band-wise denoising methods ignore spectral information and may also introduce artifacts and spectral distortion into HSIs.

Different ground objects have different reflectance under different wavelengths of radiation. As the wavelengths of adjacent bands of HSIs are narrow, HSIs contain abundant spectral information. As the number of bands increases, the correlation between these bands becomes more complex. Both spatial and spectral information are important to HSI denoising. To jointly utilize the information in both the spatial and spectral domains, multiband-based solutions have been introduced; these can be categorized into two main groups, namely spatial-domain-based methods and transform-domain-based methods.

A. Spatial-Domain-Based Methods

Spatial-domain-based methods focus on adopting reasonable assumptions or priors, which are well able to preserve spatial and spectral characteristics. For example, Fu *et al.* [7] exploited high correlation across spectra and non-local self-similarity over space in the noisy HSI to learn an adaptive spatial-spectral dictionary. Based on the learned dictionary, they exploited the local and non-local sparsity of the HSI to design a denoising model. Zhang *et al.* [8] explored the low-rank property of HSI, finding evidence that a clean HSI patch can be regarded as a low-rank matrix, and formulated the HSI restoration problem using a low-rank matrix recovery (LRMR) framework. In addition, the low-rank property and

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spatial piecewise smoothness of HSIs, as assumed, inspired tensor-based and total variational regularization methods. Fan *et al*. [9] proposed a spatial-spectral total variation regularized low-rank tensor factorization (SSTV-LRTF) method to remove mixed noise in HSIs; this approach simultaneously considered the local spatial structure and spectral correlation of neighboring bands and achieved superior HSI mixed-noise removal performance. Yuan *et al*. [10] applied the total variation algorithm to both the spatial and spectral views and designed a metric Q-weighted fusion algorithm to merge these two views, achieving improved denoising results.

B. Transform-Domain-Based Methods

By exploiting the sparsity of the image, transform-domain-based methods transform the image into another form and separate clear signals from the noisy data; such methods include principal component analysis (PCA), Fourier transform, and wavelet transform. For example, Othman *et al*. [11] proposed the hybrid spatial–spectral noise reduction (HSSNR) method, which considers both spatial and spectral information by using the wavelet transform of the mixed spatial-spectral derivative domain. Rasti *et al*. [12] applied first-order roughness penalty (FORP) to the wavelet domain in order to exploit the multiresolution analysis (MRA) property of wavelets to considerable effect.

Although these well-established HSI denoising methods can achieve good results, the mapping from noisy data to clean data and the image feature representation are primarily fixed functions, the parameters of which must be precisely tuned if good performance is to be attained. Furthermore, these primitive methods can also be both unstable and sensitive to the band number of HSIs. In recent years, deep learning techniques have been widely applied in the remote sensing field for a variety of tasks, including classification [13], segmentation [14], target detection [15], and data enhancement [16]. By merging the linear and nonlinear operations, data-driven deep learning methods build an automatic and hierarchical learning model from the data itself in an end-to-end fashion and are thus able to arrive at a representation that is suitable to the task at hand.

In terms of HSI denoising tasks, several typical problems, i.e., a combination of spatial and spectral information, the extraction of complex information from HSIs, and the representation of the inherent multiscale features can be solved by means of deep learning models.

To address the first problem, Yuan *et al*. [17] directly set the current spatial band and its k adjacent bands as the inputs for two parallel feature extraction branches. Zhang *et al*. [18] took the simulated kth noisy band, its horizontal/vertical spatial gradient, and its adjacent spectral gradient as inputs to learn the spatial and spectral items via a designed spatial–spectral loss function. Methods of this kind have proven that extracting the spatial and spectral information in a manner that does not create interference can effectively improve the performance of deep learning-based methods. However, most existing works use cascaded convolutional blocks or human-crafted preprocessing for parallel feature extraction, which may result in the intrinsic spatial and spectral correlation embedded in HSIs being ignored.

The second issue is that of how to construct effective feature extraction modules for HSIs. HSI denoising is a low-level feature learning task, in which block-stacked modules [19] are often used for further image feature extraction. It has been proven that most very difficult problems in the computer vision field can be tackled by a deep neural network ([20], [21]). Nevertheless, the experimental results presented in [22] showed that network depth is not the key to better image denoising results. More specifically, Liu *et al*. evaluated two deep CNN denoising networks, namely DnCNN [23] and RED-Net [24], on the gray-scale BMD100 and BMD200 data sets [25] and compared their performances to the primitive non-CNN method BM3-D. The results obtained from DnCNN and RED-Net reveal that the average elevated margins measured via peak signal-to-noise ratio (PSNR) did not even reach 1 dB, while the denoising quality also degraded as the data set was enlarged. It is true that deep CNNs can model the high-level semantic information using a series of linear and nonlinear layers. As the network depth increases, the low-middle-high level features are increasingly integrated . However, HSI denoising is a relatively low-level task and requires more pixel-level features, meaning that a deep network may not be suitable for the task.

Thirdly, the multiscale feature expression of ground objects has always been key to enhancing the performance of computer vision tasks. For one thing, the diverse object sizes, varying distances from the imaging sensors, and other factors collectively lead to different sizes of objects presented in the image. For another, the features of small-scale objects in high-level layers are usually inconspicuous and can even be omitted. A large number of multiscale strategies have been proposed to resolve the above-mentioned issues. Commonly used methods include employing convolution with different kernel sizes or highlighting low-level features with high resolution. Different kernel sizes introduce a series of receptive fields, i.e., the kernel scans the feature map and captures the local contextual information of the neighborhood as defined in a variety of field of views. Similar strategies have been explored in the field of HSI denoising. In HSID-CNN [17], the model has introduced a multiscale convolutional unit to extract the multicontext of both the spatial and spectral features. In more detail, the convolution kernel sizes are set to be 3×3, 5×5, and 7×7 respectively. Moreover, all output layers are concatenated together to generate the joint spatial-spectral multiscale features. While plain large 2-D kernels can enlarge the receptive field, this can also lead to an increase in computational cost.

In this article, we propose a discriminative 3-D attention denoising network (3-D-ADNet). In the proposed model, we consider the correlation between adjacent bands and take an individual band and its adjacent multiband as inputs. Corresponding to the inputs, the information extraction is divided into a spatial branch and a spatial-spectral branch, which employ 2-D convolution and 3-D atrous convolution respectively. A self-attention mechanism is applied to both the

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spatial and spectral domain before the branches are combined, which enables the local features to be adaptively and dynamically integrated with their global dependencies. The dual attention modules can enhance the feature representation capability and provide a foundation for better HSI denoising and restoration results. We then concatenate the spatial and spectral features and utilize three 3-D convolution layers to extract large features, an approach that reduces feature redundancy and effectively integrates the spatial and spectral features. Subsequently, we design four parallel 2-D convolution layers with different dilation rates to extract multiscale features. The atrous convolution for multiscale feature fusion can effectively enlarge the receptive fields and reduce the computational burden with fewer parameters. The main contributions of this work can be summarized as follows.

1) We apply a self-attention mechanism to both the spatial and spectral domains before the branches are combined. The self-attention mechanism can use the input feature itself in order to formulate this feature's importance to the result, thus yielding a performance improvement. For the spatial domain, the attention module can formulate interdependencies on the feature map. For the spectral domain, the attention module can simulate the spectral correlation well.
2) A lightweight strategy is used for large-scale feature and multiscale structure extraction. Three 3-D convolution blocks are designed to extract large-scale features and 2-D atrous convolutions, with different dilation rates used to form a multiscale structure. With a reduced computational burden, both of these structures provide desirable receptive fields for different sizes of ground objects and different numbers of image bands.
3) Our proposed method can deal with noisy images with varying noise intensity and can also effectively cope with real scene noise removal without the need for parameter-setting. Moreover, the performance of subsequent applications (i.e. HSI classification with SVM) can be significantly improved after noise is removed via the proposed model.

The remainder of this article is organized as follows. In Section II, related 3-D HSI denoising networks and self-attention mechanism are introduced. The framework and details of the proposed model are outlined in Section III. The simulated and real-data experimental results are reported and discussed in Section IV. Finally, Section V concludes this article.

II. Related Works

A. 3-D Convolution and 3-D HSI Denoising Networks

Standard 2-D convolution operates as follows: a learned 2-D convolution kernel moves along the spatial dimension and sums the product of kernel weight and the pixel value within the corresponding area, to capture spatial features and generate a feature map of one channel [Fig. 1(a)]. Despite the success of 2-D convolution, it is extremely ineffective for hyperspectral data and can easily lead to spectral distortion, as it corrupts the correlation among the bands at the beginning of convolution.

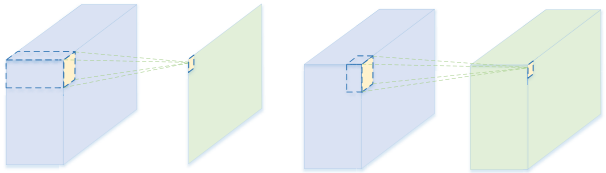


Fig. 1. Comparison between 2-D convolution and 3-D convolution. (a) 2-D convolution. (b) 3-D convolution.

Therefore, the question of how to fully utilize the abundant spectral information of HSI gives rise to the application of 3-D convolution. Unlike 2-D convolution, the 3-D convolution kernel is three-dimensional; that is, it slides on both the spatial and spectral dimensions. As displayed in Fig. 1(b), it outputs a volume stacked with many cubes, where each cube is produced by summing the weighted values of the input multiband. In this way, 3-D convolution is better able to preserve the inherent spectral information of HSI in multiple tasks. In [26], a 3-D convolutional neural network is proposed to fuse multispectral and HSIs to obtain a high-resolution HSI. In [27], the proposed method also employed 3-D convolution for accurate HSI classification, enabling the deep spatial-spectral-combined features to be effectively extracted without any preprocessing or post-processing.

The application of 3-D convolution to HSI denoising has also been explored. Liu and Lee [28] proposed a 3-D atrous denoising convolution neural network (3-D ADCNN) for HSI. Through the integration of 3-D convolution and atrous convolution, the model extracts feature maps along both the spatial and spectral dimensions and enlarges the receptive field without significantly increasing the number of network parameters. However, the network takes the whole HSI as input and uses 3-D convolution for all feature extraction blocks; this approach does not consider the differences between the spatial and spectral domains of HSI.

B. Self-Attention Mechanism

The key principle behind the machine-learning attention mechanism involves finding the most salient and important target for task solving, which is similar to the human attention mechanism. Successful applications of the attention mechanism have been documented in both natural language processing (NLP) ([29]–[31]) and computer vision tasks ([32], [33]). In particular, self-attention, also referred to as intra-attention, models the response at a particular position within a sequence by attending to all positions within the same sequence. Self-attention has been widely applied to many types of tasks. Tan *et al*. [34] presented a model for semantic role labeling (SRL) based on the self-attention mechanism, which is able to directly capture the relationships between two tokens regardless of the distance between them. Zhang *et al*. [35] explored self-attention in the GAN context. The generator developed by these authors can generate high-resolution details using cues from all feature locations, while the discriminator is able to check the consistency of distant portions of the image. In our work, the self-attention module we employ is based on the dual

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attention network (DAN) proposed by Fu *et al*. [36] for scene segmentation. The DAN captures the global context with a position attention module and channel attention module, which are built on the top of the traditional dilated FCN. The network has achieved a new state-of-the-art segmentation performance on several challenging scene segmentation data sets. In the HSI denoising task, due to the noise distribution being different in the spatial and spectral dimension, the attention mechanism was used to capture the global dependence of the spatial and spectral domains in our proposed method.

III. Methodology

In this section, we first present an overview of the proposed HSI denoising framework. To jointly utilize the inherent spatial–spectral properties of the HSI, a 3-D attention denoising network (3-D-ADNet) is designed to map the noisy HSI to clean data on a band-by-band basis. The overall network architecture is depicted in Fig. 2. The network takes the jth noisy band and its k adjacent bands as input, then feeds them into the spatial feature extraction branch and spectral feature extraction branch, respectively. The features extracted from the two branches are then fed into the dual attention module to produce long-range contextual information. Following the aggregation of spatial and spectral information, we employ consecutive 3-D convolution blocks and skip connection to generate large-scale features. Finally, the multiscale structure and 2-D convolution are employed to facilitate the better restoration of the HSI. The network is trained with noisy data and the corresponding clean data, and weights are updated using the back-propagation (BP) algorithm. After the loss is converged, we can obtain the clean HSI by going through all bands of the HSI in this way.

HSI denoising is the process of separating the clean image from the noise, and can be simplified as follows:

$$\mathbf{Y} = \mathbf{X} + \mathbf{N} \tag{1}$$

where $\mathbf{Y} \in \mathbb{R}^{C\times H\times W}$ is the original noisy image, while $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$ and $\mathbf{N} \in \mathbb{R}^{C\times H\times W}$ denote the clean image and the noise with Gaussian distribution respectively; moreover, C refers to the number of bands, while H and W refer to the height and width respectively. As can be seen from Fig. 2, residual learning is employed in our proposed model. The output of the model is a residual image; subsequently, we adopt a skip connection that adds the predicted residual noise $\hat{\mathbf{N}}$ to the original noisy image, which can be defined as follows:

$$\mathbf{Y} = \hat{\mathbf{X}} + \hat{N} \tag{2}$$

where $\hat{\mathbf{Y}}$ is the predicted image. The mean-squared error between the predicted image $\hat{\mathbf{Y}}$ and the clean image $\mathbf{X}$ is employed as the loss function $\mathcal{L}(\Theta)$ of our network

$$\mathcal{L}(\Theta) = \frac{1}{2N}\sum_{i=1}^{N} \left\| \mathcal{F}\left(\mathbf{Y}_i, \mathbf{Y}_i{}^k, \Theta\right) - \mathbf{N}_i \right\|^2 \tag{3}$$

where Θ denotes the trainable parameters, $\mathbf{Y}_i$ and $\mathbf{Y}_i{}^k$ refer to the spatial input and the spectral input respectively, and $\mathbf{N}_i$ is the residual noise.

A. Spatial-Spectral Information Extraction

The spatial–spectral information extraction module comprises the spatial processing branch and the spectral processing branch.

Both the spatial and spectral information extraction is carried out by stacking four convolution blocks; each block is composed of the convolution, batch normalization (BN), and leaky rectified linear unit (LeakyReLU). For the spatial processing branch, we use 2-D convolution to capture the local dependencies among neighboring pixels. We employ 3×3 kernels and set the dilation rate for each level to [1, 1, 3, 5], meaning that only the last two blocks apply atrous convolution. The output channels of the convolution blocks are [k, 32, 64, 64] respectively, where k denotes the number of adjacent bands mentioned above. The size of the single band input is assumed to be $H \times W$, where H and W represent the height and the width respectively. The output size of the spatial process branch will be $k \times H \times W$, which is the same as the input of adjacent bands, in order to prepare for the spatial–spectral information fusion.

However, the spectral processing branch employs 3-D convolution in the convolution blocks. The 3-D kernel can enlarge the receptive fields in spectral bands by moving along the spectral dimension on one hand; on the other hand, it can model the strong spectral correlation within a specific and continuous spectral scope, as specified by the channel dimension of the 3-D kernel. Simultaneously, the utilization of dilated kernels ensures that spatial receptive fields are larger without increasing the number of parameters. The size of the 3-D convolution output volume is automatically determined by the size of the input feature and the size of the convolution kernel. Despite the fact that the number of input feature channels may change in different situations, 3-D convolution will output cubes as the same number as kernels, ensuring its breadth and applicability. In our work, we set the size of the 3-D kernel as $5 \times 3 \times 3$, while the dilation rates are [1, 1, 3, 5]. The output volume numbers for each 3-D convolution in the consecutive blocks are [4, 8, 8, 1], A cube with size $k \times H \times W$ is output as the spectral feature representation.

B. Spatial-Spectral Feature Fusion

After the spatial and spectral features have been obtained, they are fed into the dual attention module; namely, the position attention module and channel attention module, respectively. A more detailed structure is presented in Fig. 3. Due to the limitation on the size of the convolution kernel, traditional convolution operations can only obtain local receptive fields; however, the distribution of noise usually has global characteristics [37]. To break the image quality improvement bottleneck, the position attention module is introduced to capture the global relationships among different positions on the feature map. As can be seen from Fig. 3(a), the spatial feature $\mathbf{F}_{\text{spat}} \in \mathbb{R}^{C\times H\times W}$ is first fed into different convolution layers, followed by BN and LeakyReLU layers, which generates three feature maps $\mathbf{F}_1$,$\mathbf{F}_2$,$\mathbf{F}_3$, where $\{\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3\} \in \mathbb{R}^{C\times H\times W}$ and C denotes the number of channels of the feature. The new features are then all reshaped to $\mathbb{R}^{C\times N}$, where N is equal to the number of

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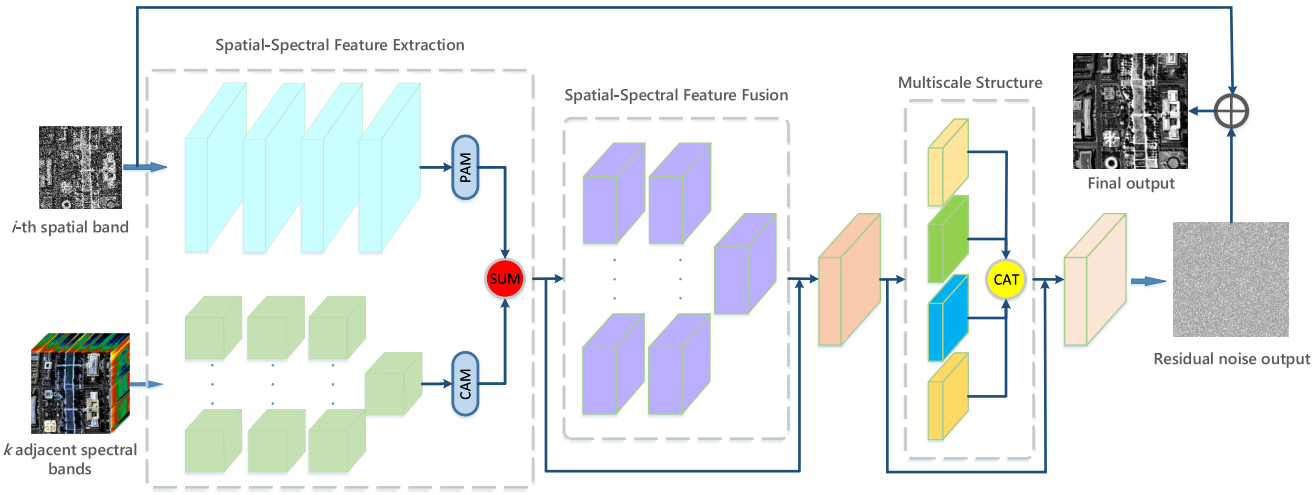


Fig. 2. Structure of 3-D-ADNet.

pixels in the feature map. To obtain the spatial attention map $\mathbf{A}_{\text{spat}}$, we multiply the transpose of $\mathbf{F}_1$ with $\mathbf{F}_2$ and pass the result to a softmax layer. The matrix multiplication calculates the product of any two pixel values, while the softmax function normalizes all feature values with respect to every position; therefore, the a_{ij} on the attention map $\mathbf{A}_{\text{spat}} \in \mathbb{R}^{N\times N}$ represents the weighted influence of the jth position on the ith position on the original feature map. Subsequently, we project the attention map to the spatial feature domain by implementing the matrix multiplication between the spatial feature $\mathbf{F}_3$ and the transpose of the spatial attention map $\mathbf{A}_{spat}$ and reshape the result $\mathbf{F}_4 \in \mathbb{R}^{C\times N}$ to the original feature size $\mathbb{R}^{C\times H\times W}$. Finally, the final output with long-range contextual representation is obtained by multiplying $\mathbf{F}_4$ by a scale coefficient α and adding it to the input feature $\mathbf{F}_{spat}$ in an element-wise manner. α is initialized as 0 and gradually learned throughout the training process.

In addition to the spatial feature dependencies among different positions, different channels of the spectral feature also relate to each other. In terms of the HSI spectra, each specific ground object has a distinctive kind of spectral curve [38]; this is also the case in the spectral feature space. The extreme points and the direction of the curve are fixed within a certain spectral range, i.e. any two channels of the spectral feature have correlations between them. For example, the feature representation is similar in adjacent channels. We therefore employ the channel attention module to emphasize the interdependencies between feature channel maps, as illustrated in Fig. 3(b). Unlike in the position channel module, we obtain the spectral feature map by directly using the input map $\mathbf{F}_{\text{spec}}$, since this enables the relationships between different channel maps to be maintained [36]. The feature matrix $\mathbf{F}_{\text{spec}}$ is then reshaped to $\mathbb{R}^{C\times N}$. We perform a matrix multiplication between the reshaped $\mathbf{F}_{\text{spec}}$ and its transpose, then apply a softmax layer to obtain the spectral attention map $\mathbf{A}_{\text{spec}} \in \mathbb{R}^{C\times C}$. For the multiplication, the sum of the product for the corresponding pixel values in every two channels is calculated. Subsequently, the transpose of the spectral attention map $\mathbf{A}_{spec}$ is multiplied by the reshaped $\mathbf{F}_{spec}$. We then multiply the result

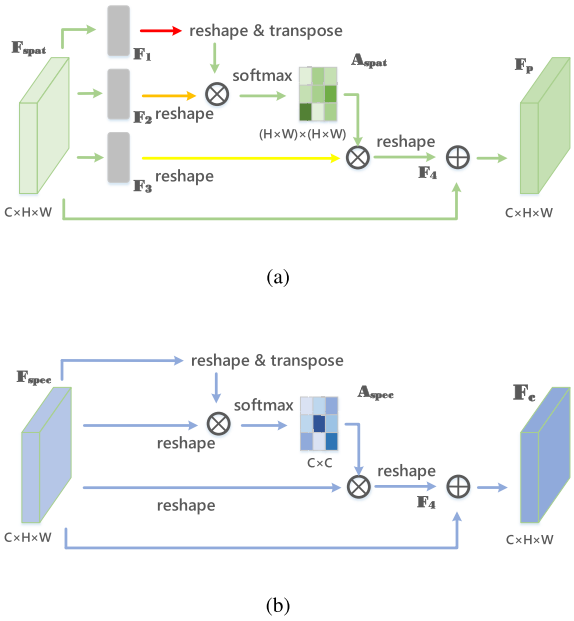


Fig. 3. Details of the position attention module (a) and the channel attention module (b).

$\mathbf{F}_4$ by a scale coefficient β and add it to the original map $\mathbf{F}_0$ to obtain the final output, where β is initially set to 0. The entire process can be expressed as follows:

$$\mathbf{F}_{\text{out}} = \{\mathbf{F}_{\text{spat}} + \alpha\text{PAM}(\mathbf{F}_{\text{spat}})\} + \{\mathbf{F}_{\text{spec}} + \beta\text{CAM}(\mathbf{F}_{\text{spec}})\}. \quad (4)$$

Here, PAM$(*)$ is the position attention module, CAM$(*)$ is the channel attention module, and $\mathbf{F}_{\text{out}}$ is the fusion of the spatial-spectral feature.

C. Large-Scale Feature Extraction

After the combination of spatial and spectral information has been performed, further feature extraction is required in order to integrate them properly. In our work, we design three convolution blocks to build a short path and employ 3-D convolution to extract the large-scale pixel-level features. The

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TABLE I
QUANTITATIVE ASSESSMENTS OF DIFFERENT METHODS UNDER DIFFERENT NOISE LEVELS WITH $\sigma = 5$ TO 100

σ	Index	Noisy	BWBM3D	KSVD	NLM3D	BM4D	ITSReg	HSI-DeNet	HSID-CNN	3D-ADNet
5	MPSNR	34.15	35.19	35.61	39.79	41.17	38.04	40.69	41.70	**42.08**
	MSSIM	0.9816	0.9871	0.9874	0.9928	0.9961	0.9943	0.9960	0.9966	**0.9968**
	MFSIM	0.9833	0.9883	0.9873	0.9944	0.9963	0.9951	0.9957	0.9965	**0.9968**
	ERGAS	50.77	45.00	42.99	31.36	24.21	33.47	25.17	23.14	**22.06**
	MSAM	0.0914	0.0752	0.0785	0.0525	0.0442	0.0411	0.045	0.0406	**0.0382**
25	MPSNR	20.17	24.54	26.64	26.57	31.10	33.01	31.83	33.28	**33.78**
	MSSIM	0.7553	0.8796	0.9163	0.8776	0.9675	0.9813	0.9753	0.9801	**0.9825**
	MFSIM	0.8650	0.9216	0.9435	0.9262	0.9737	0.9837	0.9788	0.9823	**0.9850**
	ERGAS	253.94	153.12	120.92	123.00	73.17	59.31	67.95	58.29	**54.48**
	MSAM	0.3357	0.1713	0.1518	0.1207	0.1077	**0.0736**	0.0985	0.0870	0.0804
50	MPSNR	14.16	20.80	21.70	20.79	26.77	29.50	28.34	29.11	**29.73**
	MSSIM	0.4999	0.7373	0.7514	0.6837	0.9180	**0.9599**	0.9459	0.9541	0.9587
	MFSIM	0.7531	0.8503	0.8610	0.7996	0.9429	0.9662	0.9597	0.9648	**0.9677**
	ERGAS	507.54	235.31	212.53	235.93	119.22	87.56	99.38	92.63	**85.02**
	MSAM	0.5504	0.2074	0.1875	0.1599	0.1477	**0.1016**	0.1336	0.1209	0.1112
75	MPSNR	10.63	18.98	18.86	18.86	24.29	27.26	25.68	26.98	**27.35**
	MSSIM	0.3358	0.6242	0.5694	0.5705	0.8625	**0.9366**	0.9127	0.9268	0.9326
	MFSIM	0.6682	0.7904	0.7538	0.7259	0.9121	0.9487	0.939	0.9474	**0.9510**
	ERGAS	761.35	290.51	294.27	294.33	158.10	112.83	134.4	116.61	**111.42**
	MSAM	0.7078	0.2371	0.2119	0.1825	0.1764	**0.1232**	0.1739	0.1447	0.1361
100	MPSNR	8.13	17.81	17.08	18.02	22.59	23.94	24.54	25.21	**25.74**
	MSSIM	0.2330	0.5368	0.4245	0.5098	0.8055	0.8731	0.8825	0.8992	**0.9064**
	MFSIM	0.6009	0.7434	0.6559	0.6923	0.8825	0.9157	0.9235	0.9325	**0.9358**
	ERGAS	1015.77	332.25	360.81	324.42	191.92	164.50	154.14	140.84	**133.48**
	MSAM	0.8273	0.2612	0.2294	0.1976	0.1981	**0.1392**	0.1885	0.1645	0.1534

kernel size is $5 \times 5 \times 5$ and the numbers of output blocks are [8, 8, 1].

Implementing concatenation as a skip connection is widely adopted in many well-established networks. For example, U-Net [39] concatenates the encoder and decoder features with the same resolution in the decoding process. GoogLeNet ([21], [40]), also known as Inception, also utilizes concatenation to integrate multiscale features in every Inception block. Moreover, Huang *et al*. [41] proposed the densely connected convolutional network (DenseNet) by concatenating all preceding layers to the new ones. A dense connectivity pattern of this kind enables all layers to spread their weights within or among the blocks and can achieve remarkable results more simply and efficiently. In our large-scale feature extraction module, we concatenate the module's input and output for feature reuse, which makes the model more stable and efficient. Since the input feature is generated by a dual attention module and contains rich contextual information, direct concatenation can preserve the global context inside every channel more effectively than element-wise summation, which will disturb the regularity of the original input features. Following the skip connection, the number of output feature channels reaches $2k$. We employ a 2-D convolution to halve the channels to k to lessen the parameters of the subsequent process with a kernel size of 3×3.

D. Multiscale Structure

Due to the fact that primitive multiscale structures employ plain large 2-D kernels and thus increase the associated computational costs, we embed the atrous convolution [42] in our multiscale feature extraction, using k-kernels with dilation rate to enlarge the receptive fields without the need for extra parameters. Inspired by the atrous spatial pyramid pooling (ASPP) structure [43], we design four parallel atrous convolution-BN-LeakyReLU blocks, with dilation rates as [1, 3, 5, 7] respectively, meaning that the first convolution layer is a standard convolution layer (Fig. 4). All convolution kernel sizes are set to 3×3, while all the output number of channels are set to k. Designs of this kind can reduce the calculation complexity while simultaneously maintaining diverse receptive fields. The four output features of different scales are then concatenated. Meanwhile, to ensure better stabilization of the model training process, we employ a skip connection by concatenating the input and output of the multiscale structure. This concatenation result is then fed into a standard 2-D convolution layer followed by BN and LeakyReLU, which produces a feature map of 32 channels. Finally, a 2-D convolution layer is employed to obtain the denoising result of the one band image of HSI.

IV. EXPERIMENTAL RESULTS AND DISCUSSIONS

To validate the effectiveness of the proposed method, we trained the proposed network using the Washington DC to train the model separately and data set. In addition, to verify the performance of the proposed network, we added different noise levels compared to the results with traditional methods. After network training, we checked the performance using the subsets of the Washington DC data set.

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TABLE II
QUANTITATIVE ASSESSMENTS OF DIFFERENT METHODS UNDER DIFFERENT NOISE LEVELS WITH σ = RAND(25)

σ	Index	Noisy	BWBM3D	KSVD	NLM3D	BM4D	ITSReg	HSI-DeNet	HSID-CNN	3D-ADNet
rand(25)	MPSNR	29.14	28.21	30.50	33.11	34.46	34.87	35.99	37.22	**39.13**
	MSSIM	0.9034	0.9278	0.9552	0.9646	0.9824	0.9868	0.988	0.9901	**0.9930**
	MFSIM	0.9393	0.9549	0.9637	0.9768	0.9847	0.9883	0.9889	0.9906	**0.9936**
	ERGAS	143.21	116.39	89.38	62.52	55.10	48.78	45.30	40.79	**34.52**
	MSAM	0.2150	0.1616	0.1412	0.0874	0.0912	0.0666	0.0757	0.0666	**0.0567**

TABLE III
QUANTITATIVE ASSESSMENTS OF DIFFERENT METHODS UNDER DIFFERENT NOISE LEVELS WITH σ = GAU(200,30)

σ	Index	Noisy	BWBM3D	KSVD	NLM3D	BM4D	ITSReg	HSI-DeNet	HSID-CNN	3D-ADNet
$Gau(200, 30)$	MPSNR	28.20	27.92	28.75	34.25	33.68	35.01	36.65	37.40	**38.47**
	MSSIM	0.9044	0.9259	0.9351	0.9593	0.9619	0.9870	0.9887	0.9896	**0.9918**
	FSIM	0.9393	0.9529	0.9546	0.9736	0.9729	0.9888	0.9888	0.9904	**0.9924**
	ERGAS	136.99	114.13	104.51	67.26	79.10	47.67	42.70	40.70	**35.93**
	MSAM	0.2190	0.1676	0.1669	0.0944	0.1398	0.0650	0.0706	0.0669	**0.0587**

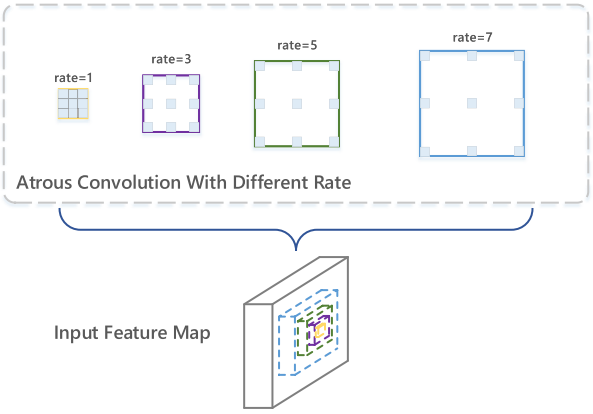


Fig. 4. Structure of 3-D-ADNet.

A. Training Data Sets and Parameter Setting

The Washington DC Mall image obtained by the Hyperspectral Digital Imagery Collection Experiment airborne sensor, with a size of 1280×303×191, was divided into two parts. The part of size of 200 × 200 × 191 is for testing, while the other part of size 1080 × 303 × 191 is for training. The training data was then cropped in a patch size of 40 × 40, with a row stride equal to 38 and column stride equal to 40, resulting in a total of 41 256 patches. The simulated noisy patches are generated through the imposition of additive white Gaussian noise on different bands. The mean of the noise distribution is 0 and the variance is σ; here, σ denotes the noise level or noise intensity. The noise intensity is multiple and has a fixed or random probability distribution for different experiments. In order to enhance the network's denoising ability, data augmentation was carried out via multiangle image rotation (angles of 0°, 90°, 180°, and 270°) during the training procedure.

In order to more effectively verify the model's robustness to different noise levels, we designed three ways to add noise.

1) Adding noise with the same intensity for different bands. For different bands, the noise intensity is equal. For example, σ is set to be from 5 to 100, as listed in Table I.
2) Adding noise with random intensity for different bands. For different bands, the noise intensity is different and conforms to a random probability distribution (as shown in Table II "σ = rand(25)" (here, rand(25) means 25 ∗ rand() in MATLAB.).
3) Add noise with Gaussian distribution for different bands. For different bands, the noise intensity is also different; here, the noise level σ is added along the spectral dimension and is varied in a manner similar to a Gaussian curve [44]

$$\sigma = \beta \sqrt{\frac{\exp\left\{-\left(n - B/2\right)^2 \Big/ 2\eta^2\right\}}{\sum_{i=1}^{B} \exp\left\{-\left(n - B/2\right)^2 \Big/ 2\eta^2\right\}}}. \quad (5)$$

Here, the intensity of the noise is restricted by β, with η being the standard deviation for the Gaussian curve. In the simulated experiments, the noise was defined as σ = Gau(β, η), where β = 200 and η = 30 (as shown in Table III "σ = Gau(200, 30)"). In order to effectively utilize the rich spectral and spatial information contained in the HSI data, the input data of our network includes both the spatial and spectral parts; specifically, a single band of the image and its adjacent k bands respectively. In our network, k is set to 24 according to the result of HSID-CNN [17]. We choose Adam as the optimizer for network training. Here, the parameter β_1 is set to 0.9, β_2 is set to 0.999, ε is set to 1e−8, and the weight decay is set to 1e−4 in order to prevent our network from overfitting. We use mean-square error loss (MSELoss) as the loss function. Our network's initial learning rate is set to 1e−3; with the increase of epochs, it decays exponentially with an attenuation index of 0.9. The total number of epochs is 120. Unlike the previous networks, we choose LeakyReLU as the activation function. As a variation on the classic ReLU activation, LeakyReLU includes a very small slope for negative value inputs. This approach mitigates against dead neurons, as the derivative is always non-zero, which allows gradient-based learning

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to occur. The training model framework is implemented in PyTorch and accelerated on NVIDIA GTX 2080ti GPU.

B. Methods for Comparison and Evaluation Index

In this article, we select several different traditional methods for comparison: the band-wise BM3-D [5] for the 2-D extended band-wise approaches; 3-D NLM [45], BM4D [46], and 3-D cube K-SVD [47] for extended 3-D-cube-based approaches; and the intrinsic tensor sparsity regularization (ITSReg) [48]. Two deep learning-based methods were chosen for comparison, namely HSID-CNN [17] and HSI-DeNet [19]. All parameters involved in the comparison algorithms have been either optimally assigned or chosen automatically as described in the references.

To evaluate the test result, five widely used indexes are chosen in our article: peak signal-to-noise ratio (PSNR), structural similarity index measurement (SSIM) [49], feature similarity index measurement (FSIM) [50], ERGAS [51], and spectral angle map (SAM) [52]. These indicators have been widely used to assess HSI denoising performance. The PSNR can be defined as follows:

$$\mathrm{PSNR} = 20\log_{10}\left(\frac{\mathrm{MAX_I}}{\sqrt{\mathrm{MSE}}}\right). \tag{6}$$

Here, MAX_I is the maximal pixel value of image $\mathbf{I}$, while MSE denotes the mean square error of the reference image and denoising image.

The SSIM between the denoising image $\mathbf{Y}$ and its ground truth $\mathbf{X}$ is defined as follows:

$$\mathrm{SSIM} = \frac{(2\mu_X\mu_Y + c_1)(2\sigma_{XY} + c_2)}{\left(\mu_X^2 + \mu_Y^2 + c_1\right)\left(\sigma_X^2 + \sigma_Y^2 + c_2\right)}. \tag{7}$$

Here, $\mathbf{X}$ and $\mathbf{Y}$ represent the original image and the predicted image, respectively; moreover, μ_X and μ_Y denote the mean of $\mathbf{X}$ and $\mathbf{Y}$ respectively, σ_X^2 and σ_Y^2 are the variance of $\mathbf{X}$ and $\mathbf{Y}$ respectively, σ_{XY} denotes the covariance of $\mathbf{X}$ and $\mathbf{Y}$, and c_2 and c_2 are constants and set to 6.5025 and 58.5225 respectively.

The FSIM can achieve very high consistency with human subjective evaluations and is defined as follows:

$$\mathrm{FSIM} = \frac{\sum_{x\in\Omega} \mathrm{S}_L(x)\mathrm{PC}_m(x)}{\sum_{x\in\Omega} \mathrm{PC}_m(x)} \tag{8}$$

where $\mathrm{S}_L(x)$ can be defined as $\mathrm{S}_L(x) = [\mathrm{S_{PC}}(x)]^\alpha[\mathrm{S}_G(x)]^\beta$, $\mathrm{S}_{PC}(x)$ represents the similarity of the phase congruency and $\mathrm{S}_G(x)$ denotes the similarity of gradient magnitude, and $\mathrm{PC}_m(x) = \max[\mathrm{PC}_1(x), \mathrm{PC}_2(x)]$.

The ERGAS provides a global quality evaluation of the denoising result and can be defined as follows:

$$\mathrm{ERGAS} = 100\frac{h}{l}\sqrt{\frac{1}{N}\sum_{i=1}^{N}[\mathrm{RMSE}(i)/\mathrm{Mean}(i)]} \tag{9}$$

where ERGAS is the ratio of the spatial resolution of the two images, i denotes the number of bands of the denoising image, $Mean(i)$ represents the mean value of differences between the i-th band of the reference image and that of the denoising image, and $RMSE(i)$ indicates the root mean squared error of the i-th band between the reference and denoising images.

The SAM measures the spectral angle between two vectors. The SAM between two vectors can be defined as follows:

$$\mathrm{SAM} = \mathrm{acos}\left(\frac{\langle \mathbf{v}, \hat{\mathbf{v}}\rangle}{\|\mathbf{v}\|_2 \cdot \|\hat{\mathbf{v}}\|_2}\right) \tag{10}$$

where v is the pixel vector of the reference image, while $\hat{v}$ is the pixel vector of the denoising image. SAM is first calculated on a per-pixel basis, after which all SAM values are averaged to a single value for the whole image.

According to the formula used to calculate the above indexes, the larger the values of PSNR, SSIM, and FSIM, the better the denoising result; moreover, the smaller the values of ERGAS and SAM, the better the denoising result.

C. Experimental Results on the Test Data Set

The test data set, which is obtained from the Washington DC Mall image, contains 191 bands with size of 200×200. In this article, the results will be assessed from the aspects of the quantitative index and qualitative analysis: more specifically, the quantitative indexes are MPSNR, MSSIM, MFSIM, ERGAS, and MSAM. MPSNR, MSSIM, and MFSIM are the average values of the results of 191 bands, while MSAM is the average results of each spectral vector.

We compared the specific values of the five indexes; the results are presented in Tables I–III. The optimal results of each indicator are marked in bold. Table I lists the experimental results of adding the same noise intensity to each band. These results indicate that the proposed 3-D-ADNet method achieves most of the optimal results. Our proposed method can remove the noise effectively for different noise levels. The results of imposing random noise intensity on each band are presented in Table II. When the noise intensity of each band is different, the quantitative indexes show that 3-D-ADNet possesses obvious advantages compared to the traditional methods; compared to the deep learning methods HSID-CNN and HSI-DeNet, our method is clearly superior on all indicators. Considering that real HSIs are usually characterized by different noise intensities, our proposed method is better able to adapt to real scene denoising. Table III presents the results of adding noise intensity with Gaussian distribution for each band. In this case, due to the varying noise levels in each band, our proposed model still achieves excellent generalization ability. The assessment values of our method are significantly superior to those of traditional methods and are also higher than the deep learning-based methods HSID-CNN and HSI-DeNet. All the above results indicate that our method achieves good denoising results, whether for different noise levels or the same noise levels.

In order to visually compare the denoising results of different methods, we output the denoising results of each method and perform a local magnification comparison. Because the test image has a substantial number of bands, we selected one or more of them for comparison. The results of this comparison are presented in Figs. 5–7. In Fig. 5, we display the pseudocolor image with band 57, band 27, and band 17. The figure shows the denoising result when the noise intensity

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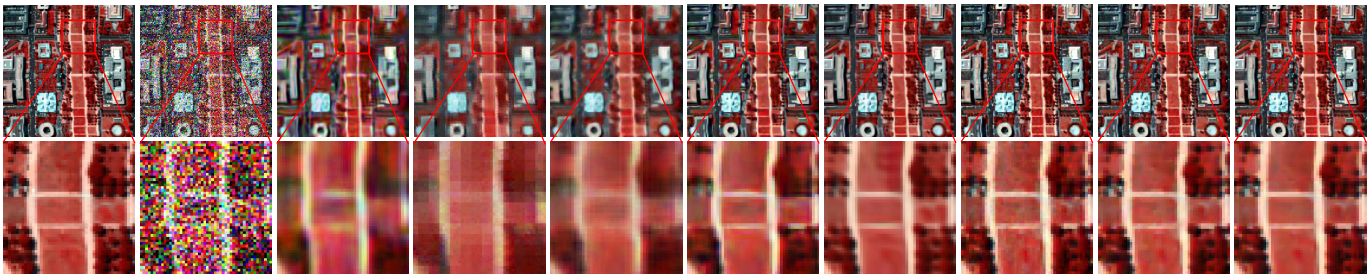


Fig. 5. Denoising results of Washington DC Mall image with $\sigma = 100$. (a) Noise-free image with band [57, 27, 17]. (b) Noisy image. (c) BWBM3-D. (d) KSVD. (e) NLM3-D. (f) BM4D. (g) ITSReg. (h) HSI-DeNet. (i) HSID-CNN. (j) 3-D-ADNet.

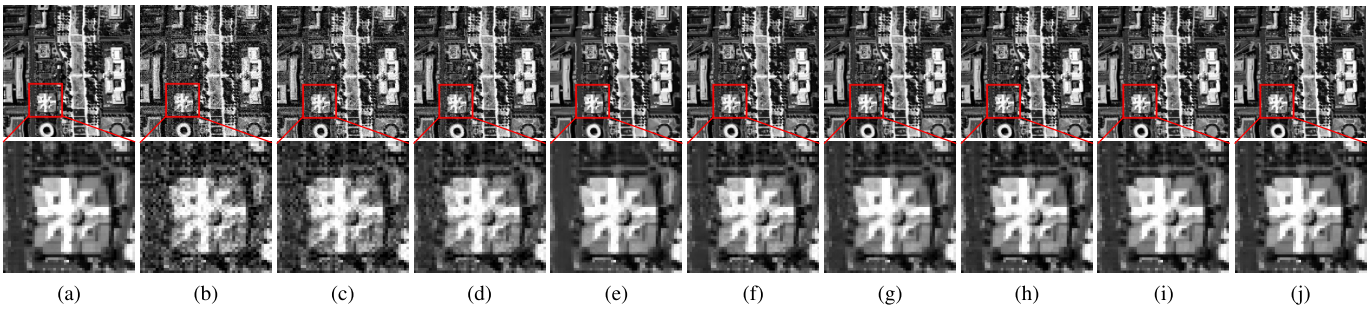


Fig. 6. Denoising results of Washington DC Mall image with $\sigma = \text{rand}(25)$. (a) Noise-free image with band 141. (b) Noisy image. (c) BWBM3-D. (d) KSVD. (e) NLM3-D. (f) BM4D. (g) ITSReg. (h) HSI-DeNet. (i) HSID-CNN. (j) 3-D-ADNet.

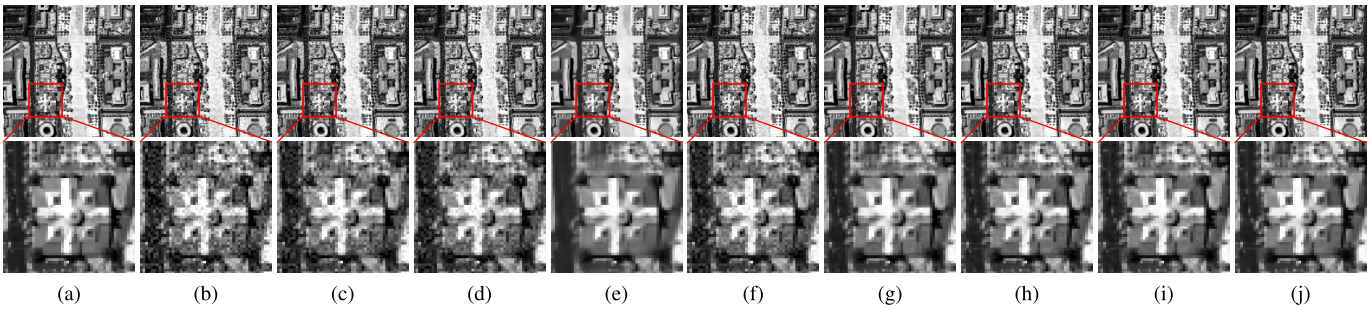


Fig. 7. Denoising results of Washington DC Mall image with $\sigma = \text{Gau}(200, 30)$. (a) Noise-free image of band 98. (b) Noisy image. (c) BWBM3-D. (d) KSVD. (e) NLM3-D. (f) BM4D. (g) ITSReg. (h) HSI-DeNet. (i) HSID-CNN. (j) 3-D-ADNet.

level is 100. We can see that the results of BM4D, ITSReg, HSID-CNN, and 3-D-ADNet are visually superior to those of other methods. Both BM4D and ITSReg exhibit over-smoothing effects, while the image details are better preserved by 3-D-ADNet, which is more in line with the original. In Fig. 6, we selected band 141 of the image for display. Fig. 6 shows the denoising result when the noise intensity is rand(25). It can be seen from the figure that BWBM3-D, KSVD and NLM3-D achieve poor noise reduction performance. By contrast, BM4D, ITSReg, HSI-DeNet, HSID-CNN, and 3-D-ADNet achieved good results in terms of visual effects. BM4D and ITSReg exhibit different degrees of distortion, while 3-D-ADNet is superior to traditional denoising algorithms in terms of both color retention and detail retention. We selected band 98 of the image for display in Fig. 7, which shows the denoising results of noise intensity levels that conform to the Gaussian distribution. The over-smoothing phenomenon is obvious for NLM3-D and ITSReg. In general, the visual effects of our proposed 3-D-ADNet under different noise conditions are superior to those of traditional denoising algorithms, while 3-D-ADNet better retains the details of HSIs, which is significant to HSI classification and unmixing.

Since we took the average results of 191 bands for PSNR, SSIM, and FSIM, these results cannot reflect the differences among every band. Accordingly, we calculated the PSNR, SSIM, and FSIM of each band separately; the results are presented in Figs. 8–10, which correspond to the results when the noise intensity level is 100, rand(25), and Gau(200, 30) respectively. The red line, representing the result of 3-D-ADNet, shows that this algorithm achieves significantly better performance than the traditional algorithms. For the noise intensity of rand(25) and Gau(200, 30), the PSNR, SSIM, and FSIM values of 3-D-ADNet on each band are higher across the board than those of traditional algorithms. For the denoising

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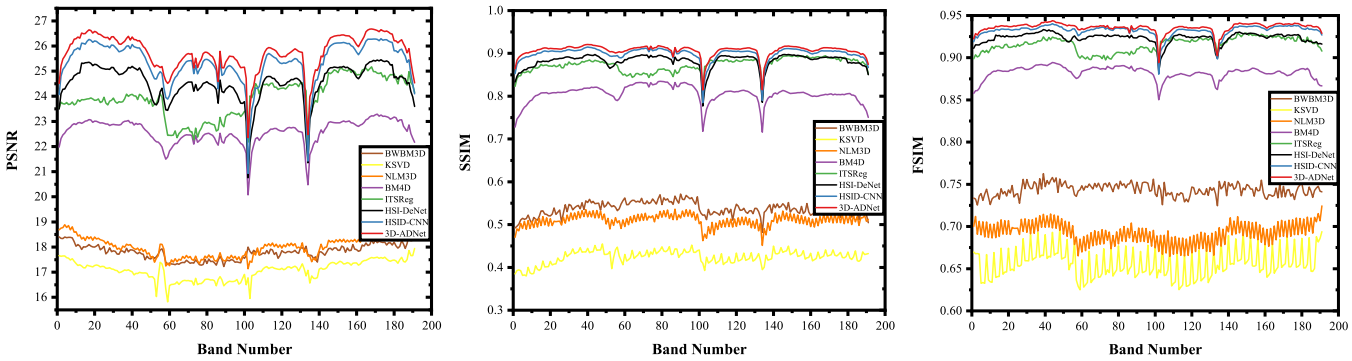


Fig. 8. PSNR, SSIM and FSIM values of different bands under different denoising methods with $\sigma = 100$. (a) PSNR values over each band. (b) SSIM values over each band. (c) FSIM values over each band.

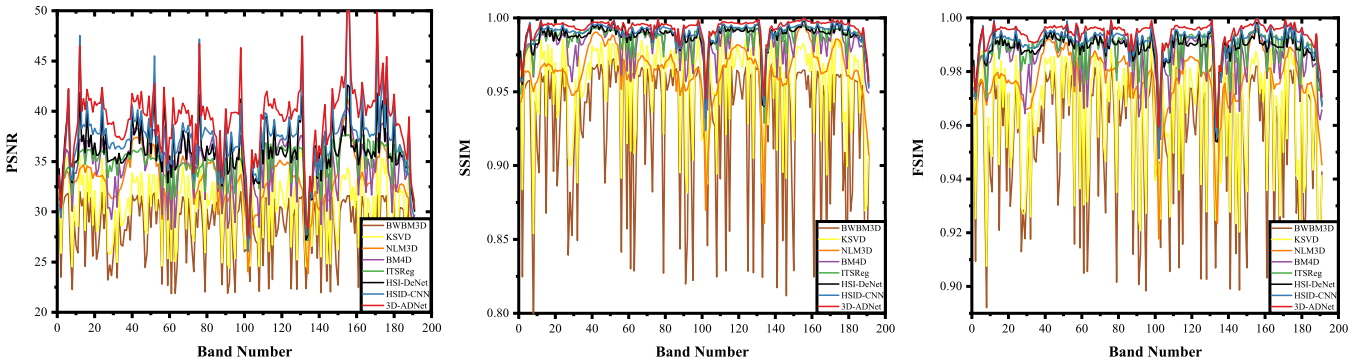


Fig. 9. PSNR, SSIM, and FSIM values of different bands under different denoising methods with $\sigma = \text{rand}(25)$. (a) PSNR values over each band. (b) SSIM values over each band. (c) FSIM values over each band.

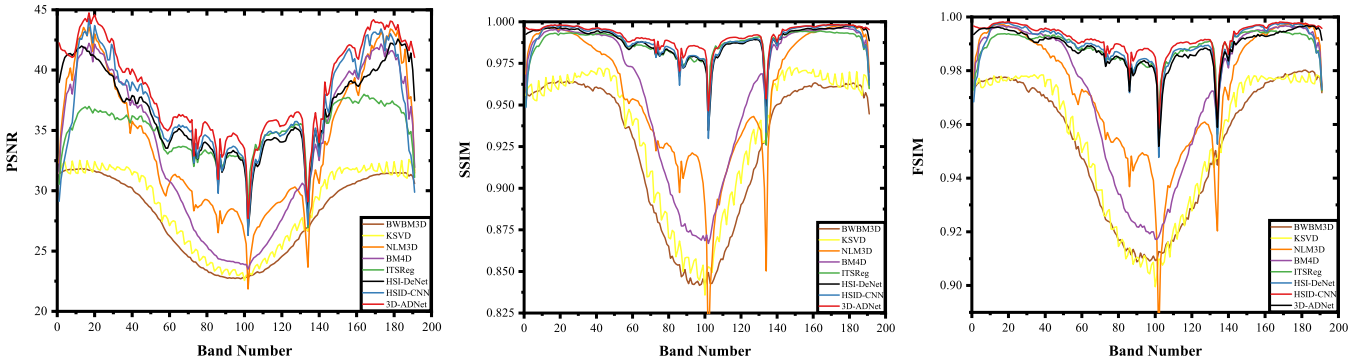


Fig. 10. PSNR, SSIM and FSIM values of different bands under different denoising methods with $\sigma = \text{Gau}(200, 30)$. (a) PSNR values over each band. (b) SSIM values over each band. (c) FSIM values over each band.

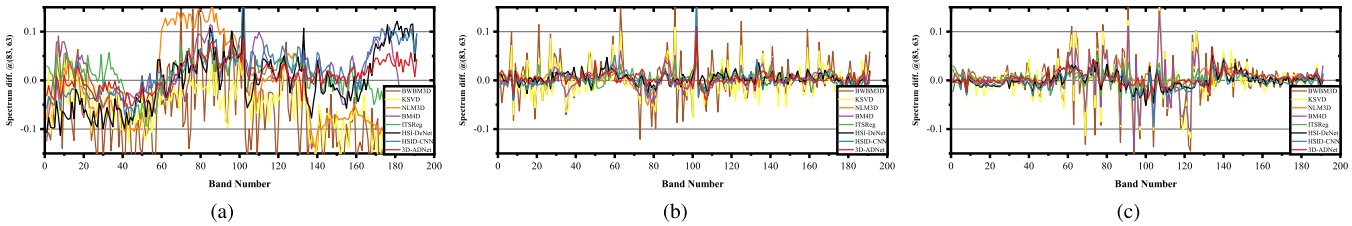
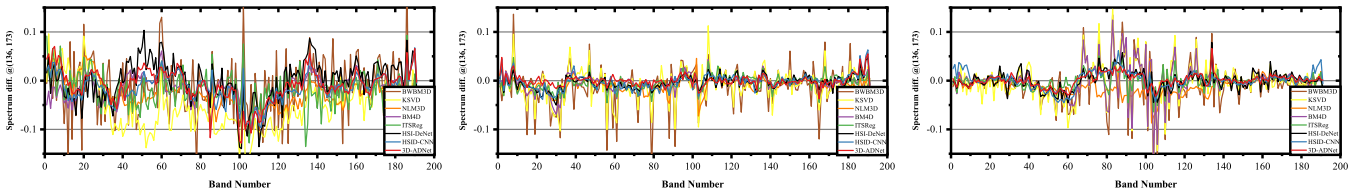


Fig. 11. Comparison of spectral reflectance difference of different denoising methods with the pixel at the location of (83, 63). (a) Spectral reflectance difference with $\sigma = 100$. (b) Spectral reflectance difference with $\sigma = \text{rand}(25)$. (c) Spectral reflectance difference with $\sigma = \text{Gau}(200, 30)$.

results after random noise processing, 3-D-ADNet can effectively avoid fluctuations caused by noise intensity changes in terms of the SSIM and FSIM indexes, while still effectively removing noise for different noise levels in different bands. Among the denoising results, the noise intensity of each band conforms to the Gaussian distribution, except for

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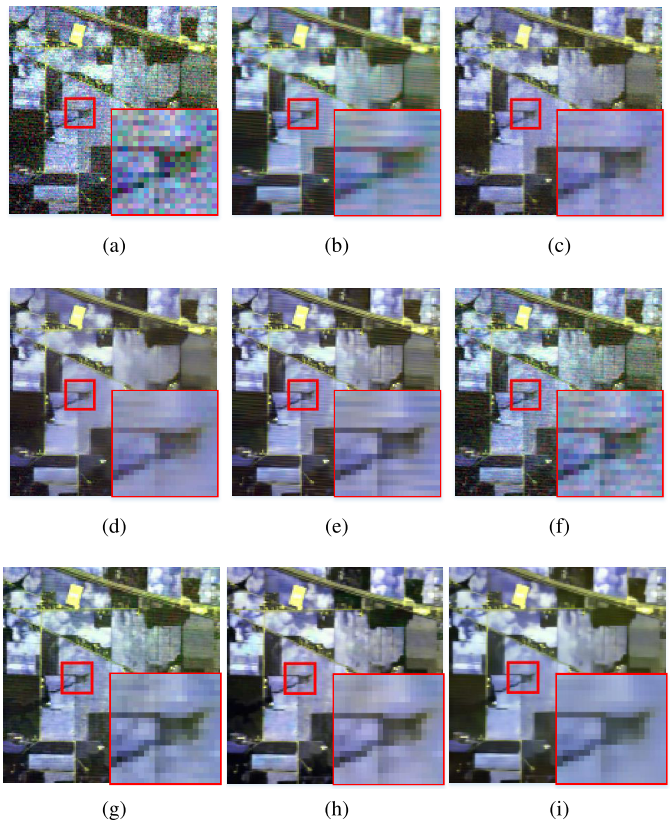
(a) (b) (c)

Fig. 12. Comparison of spectral reflectance difference of different denoising methods with the pixel at the location of (136, 173). (a) Spectral reflectance difference with $\sigma = 100$. (b) Spectral reflectance difference with $\sigma = \text{rand}(25)$. (c) Spectral reflectance difference with $\sigma = \text{Gau}(200, 30)$.

TABLE IV

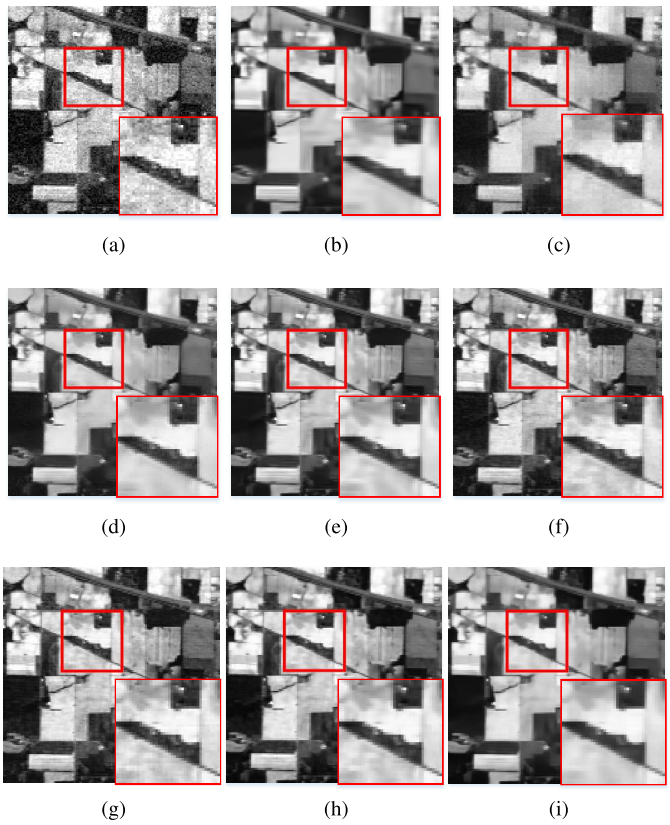
SVM CLASSIFICATION RESULTS OF THE INDIAN PINES DATA SET

Index	Original	BWBM3D	KSVD	NLM3D	BM4D	ITSReg	HSI-DeNet	HSID-CNN	3D-ADNet
OA	0.7205	0.8370	0.8279	0.8691	0.8421	0.7942	0.8520	0.8650	**0.9068**
Kappa	0.6776	0.8132	0.8028	0.8507	0.8191	0.7639	0.8306	0.8452	**0.8935**



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 13. Denoising results of the Indian Pines Data set. (a) Pseudocolor original image with bands (203, 3, 2). (b) BWBM3-D. (c) KSVD. (d) NLM3-D. (e) BM4D. (f) ITSReg. (g) HIS-DeNet. (h) HSID-CNN. (i) 3-D-ADNet.



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 14. Denoising results of the Indian Pines Data set. (a) Original image with band 202. (b) BWBM3-D. (c) KSVD. (d) NLM3-D. (e) BM4D. (f) ITSReg. (g) HIS-DeNet. (h) HSID-CNN. (i) 3-D-ADNet.

ITSReg; the other four traditional algorithms are unable to cope with images exhibiting noise intensity change, while the deep learning methods can effectively cope with this change. Of these methods, 3-D-ADNet achieves the best overall result and the fluctuation of the curve in the figure is very small. Through analysis, we determined that 3-D-ADNet has a good generalization ability and can cope with various levels of noise.

In order to better compare the spectral differences between the denoising image and the original image, we randomly select two points in the image and subtracted the spectral value of the denoising and original images; the results are shown in Figs. 11 and 12, respectively. It can be clearly seen from the figures that the results for 3-D-ADNet are small. The red curve for 3-D-ADNet is essentially distributed around 0, with a slight overall fluctuation, showing that the 3-D-ADNet denoising results can effectively maintain the original image's spectral information; this is of great significance for the subsequent application of the image.

D. Real Data Experiment

In real scenarios, HSIs usually contain noise. We accordingly selected two real hyperspectral data sets to validate the

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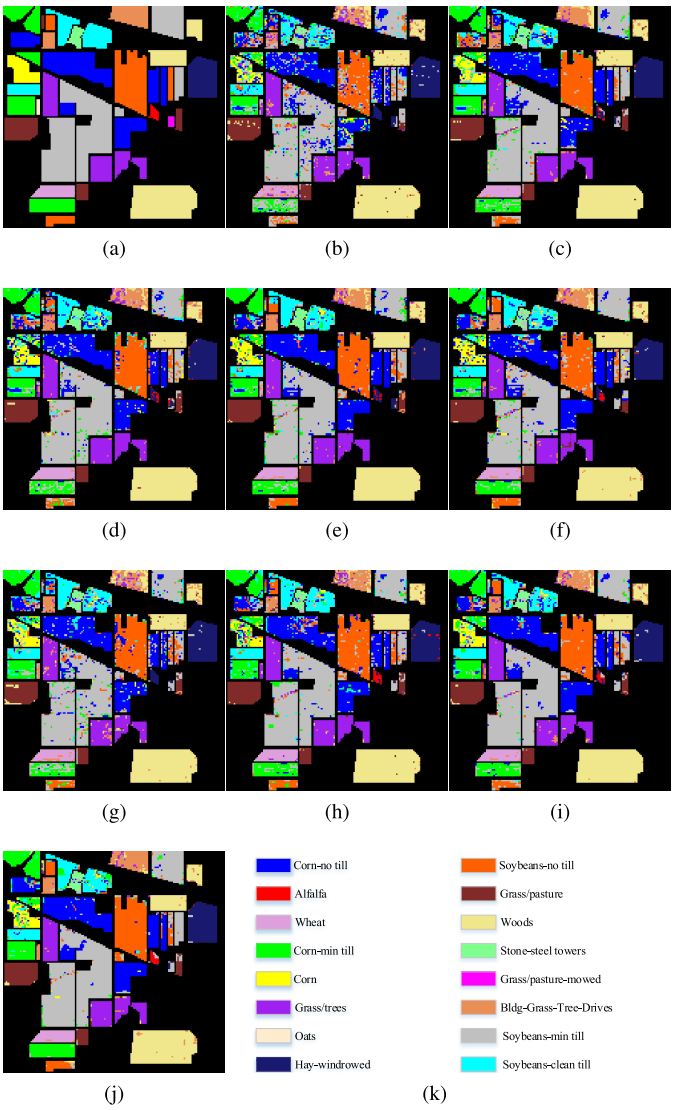


Fig. 15. SVM classification results of the Indian Pines Data set. (a) Original label. (b) Noisy image. (c) BWBM3-D. (d) KSVD. (e) NLM3-D. (f) BM4D. (g) ITSReg. (h) HSI-DeNet. (i) HSID-CNN. (j) 3-D-ADNet. (k) Legend.

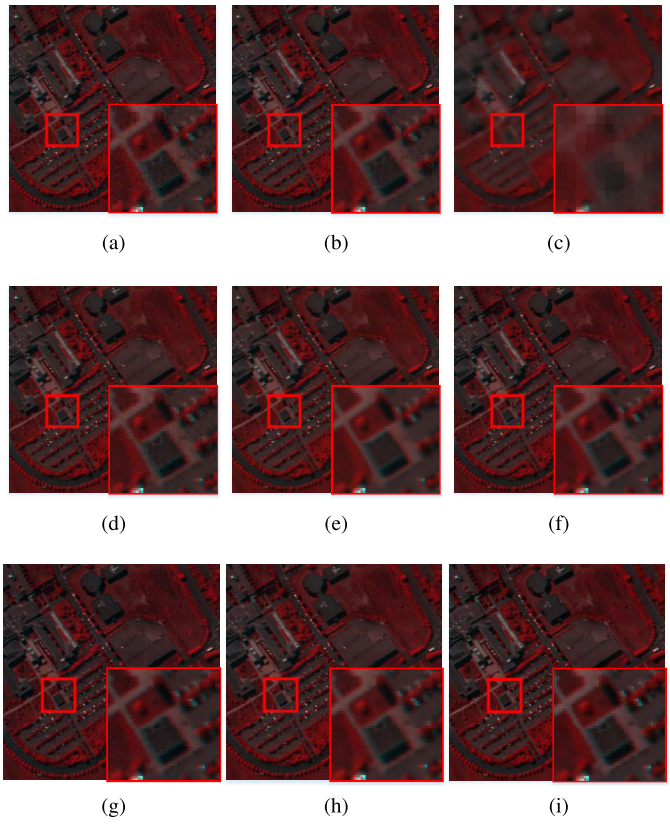


Fig. 16. Denoising results of the Pavia University Data set. (a) Pseudo-color original image with bands (203, 3, 2). (b) BWBM3-D. (c) KSVD. (d) NLM3-D. (e) BM4D. (f) ITSReg. (g) HSI-DeNet. (h) HSID-CNN. (i) 3-D-ADNet.

effectiveness of our model: the AVIRIS Indian Pines data set and the ROSIS University of Pavia data set. Both of these are widely used but contain a variety of noise. We first use different methods to remove the noise, and then classify the denoised image using SVM. Finally, we compare the visual effect and classification accuracy of the different methods. The kappa and OA (overall accuracy) are used as metrics to evaluate the accuracy of the classification results.

1) AVIRIS Indian Pines Data Set: This data set was collected by the AVIRIS sensor over the Indian Pines test site in Northwestern Indiana and consists of 145 $\times$ 145 pixels and 206 bands (after removing water absorption bands). The first and last few bands of Indian Pines are corrupted by noise; here, the noise types are mainly Gaussian and impulse noise. Figs. 13 and 14 illustrate the denoising results of the comparison algorithms and the proposed method. Fig. 13 contains pseudocolor images of combined bands (203, 3, 2), while Fig. 14 presents the results of the visual effect of the denoised 202 band.

From these results, we can clearly see that the denoising results of KSVD, ITSReg, and HSI-DeNet still contain different degrees of noise; moreover, while BWBM3-D, NLM3-D, and BM4D can remove the noise effectively, the denoising result is fuzzy overall, and some object boundaries remain unclear. Among them, BWBM3-D exhibits an obvious over-smoothing phenomenon; details and structural information of the image are accordingly lost, which is not conducive to subsequent applications. It can further be seen that HSID-CNN and 3-D-ADNet achieve good performance in terms of visual effects. In particular, 3-D-ADNet preserves both the details and boundaries of the ground objects while effectively removing noise at the same time.

Fig. 15 presents the SVM classification results following denoising by different methods. There are a total of 16 different classes of ground objects in the Indian Pines data set. We randomly selected 10% of each class to make up the training samples. The quantitative classification results are presented in Table IV. As can be seen from the table, in terms of the classification result of the original image, the OA and kappa are only 72.05% and 0.6776 . Moreover, the classification accuracy has been improved to different degrees following denoising by different algorithms. NLM3-D, HSI-DeNet, HSID-CNN, and 3-D-ADNet achieved

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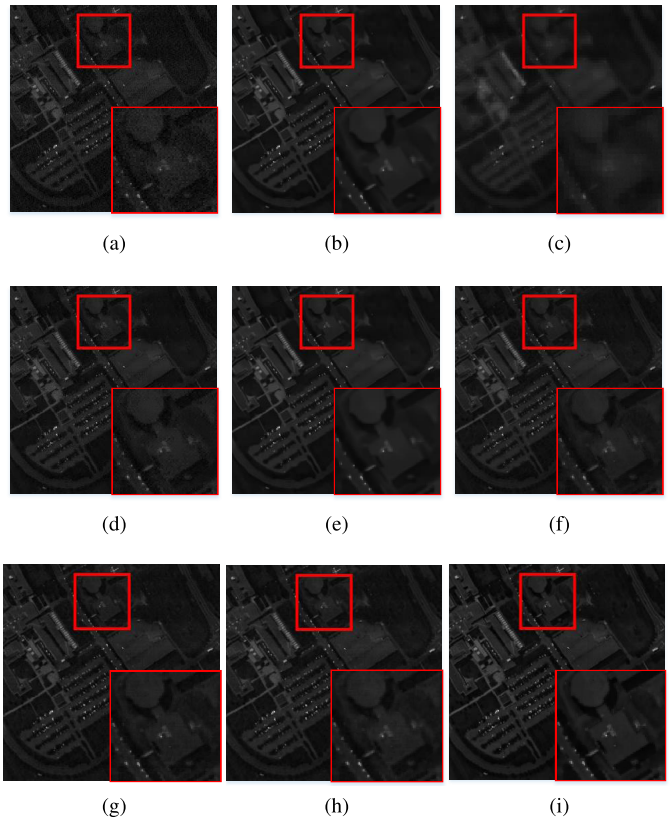


Fig. 17. Denoising results of the Pavia University Data set. (a) Original image with band 2. (b) BWBM3-D. (c) KSVD. (d) NLM3-D. (e) BM4D. (f) ITSReg. (g) HSI-DeNet. (h) HSID-CNN. (i) 3-D-ADNet.

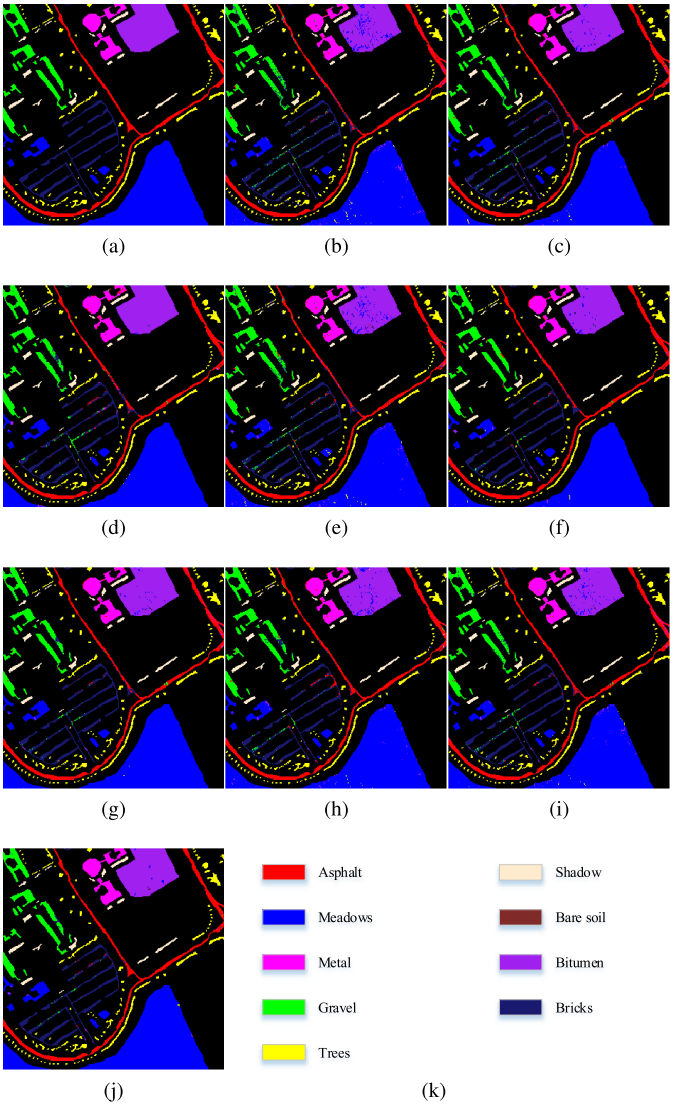


Fig. 18. SVM classification results of the Pavia University Data set. (a) Original label. (b) Noisy image. (c) BWBM3-D. (d) KSVD. (e) NLM3-D. (f) BM4D. (g) ITSReg. (h) HSI-DeNet. (i) HSID-CNN. (j) 3-D-ADNet. (k) Legend.

good results; in particular, the OA and kappa of 3-D-ADNet are as high as 90.68% and 0.8935, respectively. According to the classification map shown in Fig. 15, there are still many points misclassified by traditional methods, which shows that traditional algorithms cannot effectively cope with real scene noise removal. By contrast, 3-D-ADNet successfully overcomes this disadvantage.

2) ROSIS University of Pavia Data Set: This data set was acquired by the ROSIS sensor during a flight campaign over Pavia, northern Italy. There are 103 spectral bands and 610×340 pixels in total for Pavia University. The image ground truth differentiates between nine classes. We randomly selected the size of 300×300 for real data experiments. Most bands in the Pavia University data set are corrupted by different degrees of Gaussian noise. Figs. 16 and 17 show the original image and denoising results respectively with different band combinations. Fig. 16 contains pseudocolor images of the band (97, 3, 2), while Fig. 17 shows the results for the second band, which is seriously corrupted by Gaussian noise.

It can be observed from these figures that the denoising results of BWBM3-D and NLM3-D are barely satisfactory, as the denoised image still contains noise. Moreover, both KSVD and BM4D exhibit an obvious over-smoothing phenomenon and varying degrees of image detail loss. ITSReg, HSI-DeNet, and HSID-CNN achieve a comparatively better visual effect after denoising; however, they do not suppress the noise completely, and the structural information of the image is not well preserved. 3-D-ADNet achieves the best performance in terms of visual effect, as it removes the noise while also better preserving the details and structural information of the image.

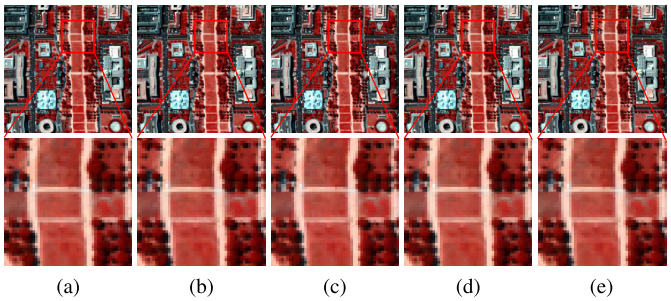


Fig. 19. Denoising results of different modules and strategy with $\sigma = \text{rand}(25)$. (a) baseline. (b) baseline + D. (c) baseline + DM. (d) baseline + DA. (e) 3-D-ADNet.

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TABLE V
SVM CLASSIFICATION RESULTS OF THE INDIAN PINES DATA SET

Index	Original	BWBM3D	KSVD	NLM3D	BM4D	ITSReg	HSI-DeNet	HSID-CNN	3D-ADNet
OA	0.9471	0.9706	0.9710	0.9515	0.9748	0.9765	0.9633	0.9651	**0.9889**
Kappa	0.9270	0.9594	0.9601	0.9330	0.9652	0.9676	0.9494	0.9518	**0.9847**

TABLE VI
INFLUENCE OF DIFFERENT MODULES AND STRATEGY WITH σ = RAND(25). (D) ATROUS CONVOLUTION. (M) MULTISCALE STRUCTURE. (A) ATTENTION MODULE

Method	PSNR	SSIM	FSIM	ERGAS	MSAM
baseline	37.23	0.9903	0.9906	40.43	0.0671
baseline+D	37.29	0.9913	0.9922	40.83	0.0655
baseline+DM	37.73	0.9917	0.9918	39.07	0.0631
baseline+DA	38.79	0.9926	0.9933	35.80	0.0594
3D-ADNet	**39.13**	**0.9930**	**0.9936**	**34.52**	**0.0567**

Fig. 18 presents the SVM classification results of the Pavia University data set, while Table V lists the quantitative results with SVM following denoising by different methods. Because only a few bands of Pavia University are corrupted by noise, it can be seen from the results that the OA and kappa of the original image are at 0.9471 and 0.9270 respectively. Nevertheless, the classification accuracy is still significantly improved after denoising. ITSReg, HSID-CNN, and 3-D-ADNet achieve high classification accuracy following denoising; of these, 3-D-ADNet achieves the best results, with OA and kappa reaching 0.9889 and 0.9847, respectively.

E. Ablation Study

This study was conducted in order to verify the effectiveness of the different modules of our proposed method. For fairness, the same parameters were used for the comparison of each component. We validated the effectiveness of the attention module (denoted as A), multiscale structure (denoted as M), and atrous convolution (denoted as D). The quantitative results of the ablation study are shown in Table V, while the denoising results are presented in Fig. 19. We can see from the table that atrous convolution slightly improves the accuracy, while the multiscale structure greatly improves the accuracy. After adding the attention module, the model accuracy improves significantly: the PSNR increases by 1.5, while the remaining indicators also exhibit significant improvement. According to these findings, our proposed model achieves the best results overall.

V. CONCLUSION

In this article, we proposed a CNN-based HSI denoising method and employed a self-attention mechanism to enhance its feature representative capacity. 3-D atrous convolution can effectively utilize the spectral information of HSIs while simultaneously increasing the receptive field; moreover, the multiscale structure extracts the multiscale features of the feature map and preserves more textural details. In addition, the trained network with residual learning can effectively exploit the high nonlinearity and obtain more essential feature extraction and expression abilities. Our results indicate that the proposed method is superior to both traditional algorithms and deep learning-based methods in both visual and quantitative evaluation. In the future, we will take some prior knowledge into consideration and design more effective modules to cope with different types of noise (e.g. strip noise and deadline noise).

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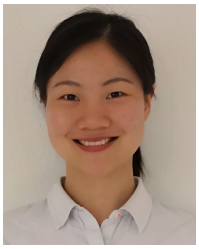
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